

Digital Agent's Engagement and Affective Posture Impact Its Social Presence and Trustworthiness in Group Decision-Making

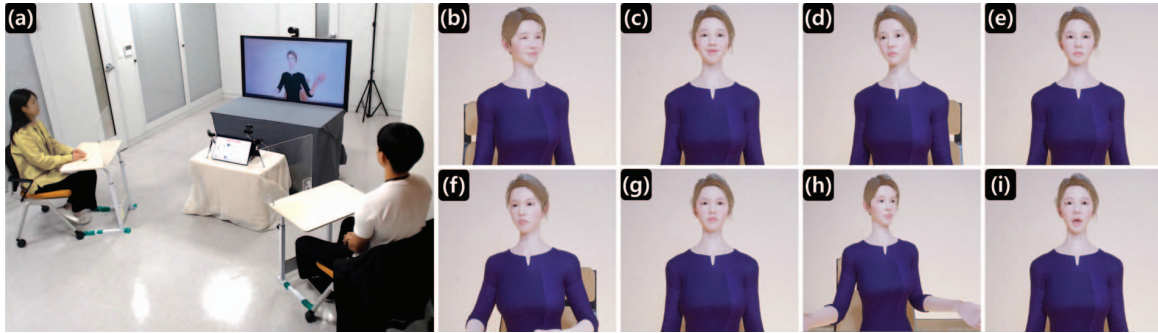
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Figure 1: Overview of our experimental materials. (a) The physical settings for a three-person group decision-making scenario with two participants and one digital agent. (b-i) the digital agent's **Engagement** and **Affective** posture as two independent variables: (b,f) E-A condition; (c,g) NE-A condition; (d,h) E-NA condition; and (e,i) NE-NA condition. During the group discussion, our digital agent is divided into two states: listening state (b-e) and speaking state (f-i).

ABSTRACT

This poster introduces a user study investigating whether a digital agent's engagement and affective posture in a group decision-making scenario influence how users perceive the digital agent's social presence and trustworthiness. Our experiment was conducted with a three-person group discussion with two participants and one digital agent, and a 2×2 within-subject study manipulating the engagement and affective posture of a digital agent. Our findings indicate that during group discussions, the engagement posture of digital agents positively affected user perception of digital agents, whereas their affective posture negatively affected user perception.

Index Terms: Human-centered computing—Human computer interaction (HCI)—HCI design and evaluation methods—User studies

1 INTRODUCTION

Many studies indicate that for users to have an ideal experience with digital agents (DAs), it is necessary for DAs to demonstrate that they are concentrating on the interaction situation at hand and/or to provide emotional feedback to users, similar to real humans [4]. However, as social beings, people often need to suppress their emotions to maintain social relationships, particularly within organizations. This is called “Expressive Suppression,” [3] and in such a situation, revealing one's emotions tends to discredit and may not be preferred as colleagues [2]. Thus, we hypothesized that DAs would have a more social presence and trustworthiness if they reduced their emotional aspects in the context of making decisions.

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In this poster, we present a human-subject study exploring how to design the DA's posture in the context of group decision-making. We designed a 2×2 within-subject study with the following two independent variables: (1) Engagement and (2) Affective posture. The engagement posture represents the DA's level of concentration on the task at hand, while affective posture conveys the DA's current emotional state. Our findings indicate that the engagement posture of DAs has a positive effect on the social presence and trustworthiness that users perceive about DAs, while the affective posture has a negative effect. The following details the experiment and the results.

2 EXPERIMENT

Physical Setup: Figure 1 (a) depicts a physical environment for group decision-making in which two participants and one DA conducted. Our DA is displayed to participants on a 55-inch monitor. We placed individuals (including one DA) seats in a triangular structure at a distance of about 2 meters. This was a precaution taken to prevent unintended situations, such as two participants forming a team and debating against the DA. Moreover, we installed seven cameras to control the experiment.

Experiment Tasks: We prepared four scenarios based on survival games to rank the importance of 15 survival items as discussion topics. We fixed the 1st, 5th, 8th, 12th, and 15th rank items. Thus, participants only need to rank 10 of the 15 available items.

Independent Variables: We devised a 2×2 within-subject study by combining the engagement (*E*) and the affective (*A*) factors: *E-A*, *E-NA*, *NE-A*, and *NE-NA*. The *E* and *NE* stated that the DA is in and not in engaging posture, respectively. Similarly, the *A* and *NA* stated that the DA is in and not in affective posture, respectively.

Digital Agent Behaviors: For each condition, we adjust various components of the DA (see Figure 1). For conditions with engagement (*E*) state, the DA will perform specific body/hand gestures while directing her gaze/body directions to the talking/listening participant. On the contrary, the DA will look upfront and not move any part of the body/hand during the non-engagement (*NE*) postures. For conditions with affective (*A*) state, the DA will express either positive or negative behavior. Precisely, the DA will express a happy

face, a positive gesture, a bright voice, and a positive utterance during the positive behavior. On the contrary, the DA will express an angry face, a negative gesture, an angry voice, and a negative utterance during negative behavior. The face, gesture, voice, and utterance of the DA will be neutral for the non-affective (NA) state.

Digital Agent's Statement and Control: We prepared 420 agreements and 420 disagreements as DA's statements. We also developed a semi-automatic system that derives the statements of the DAs upon the participant's statements. For example, if a participant states the priority of items, the experimenter will list the order of the items in the system's user interface. Our system then determines whether the items stated by the participants are correct and extracts the appropriate statements. If the participant's answer is correct, then DA expresses an agreement statement. Otherwise, DA expresses a disagreement statement. At this time, if the DA's posture is *Affective* state, agreement is expressed alongside positive behavior, and disagreement alongside negative behavior.

Experiment Procedure: One session consisted of three phases. The first phase is for participants to decide the priority of survival items alone. At this time, participants can understand the situation of the given survival game (i.e., task) and prepare for group discussion. The next phase is group discussion. The task of prioritizing item was divided into four rotations: (2nd, 3rd, 4th), (6th, 7th), (9th, 10th, 11th), and (13th, 14th). The order of speech is Person A, DA, Person B, DA, and Person A. At the end of the speech, Person B decided the priority of items. The order of participants changes with each rotation. When the four rotations are over, this phase ends.

In the last phase, a post-survey was conducted. From the prior studies [1, 5, 6], we extracted and modified several questions (see Table 1) to evaluate the user experience of the DAs.

3 RESULTS AND DISCUSSION

We recruited 32 participants (14 females, 18 males, age 22–45, $M = 27.4$, $SD = 4.9$). We used Friedman tests and Wilcoxon signed-rank tests with Bonferroni adjustment for analysis. Both were set to a level of 5% for statistical significance. Figure 2 shows the results of the user experience for the DA.

Social Presence: Significant differences were found in both continuous ($\chi^2 = 22.95$, $p < .0001$) and membership ($\chi^2 = 19.19$, $p < .001$). First, the results of continuous factor's post-hoc test are as follows: E-A < E-NA*; E-A < NE-NA*; E-NA > NE-A**; and NE-A < NE-NA** (see Figure 2 (a)). For the membership, the post-hoc test results are as follows: E-A < E-NA*; E-A < NE-NA*; E-NA > NE-A*; and NE-A < NE-NA* (see Figure 2 (b)). Based on these findings, engagement postures in such situations can marginally improve users' perceptions of digital agents as social beings, i.e., as a teammate. However, since there was no emotional suppression, the affective postures appear to have impeded the user's perception significantly. Additionally, affective posture influences user perception more than engagement posture. This may be due to the fact that the affective posture (i.e., visual/aural cues) of DAs in response to a user's statement may be more noticeable than the engagement posture (i.e., visual cues).

Trustworthiness: We found two significant effect ($\chi^2 = 12.65$, $p < .01$). Post-hoc tests showed there were two differences between E-A < E-NA*; E-A < NE-NA* (see Figure 2 (c)). Based on these results, we found that a DA's emotional behavior negatively impacts the user's perception of its trustworthiness. Particularly in the context of decision-making, emotionally restrained DAs (i.e., similar to humans) can enhance the credibility of their statements more so than emotional DAs.

4 CONCLUSION

In this poster, we conducted a user study manipulating DA's posture in the situation of group decision-making. Our findings suggested

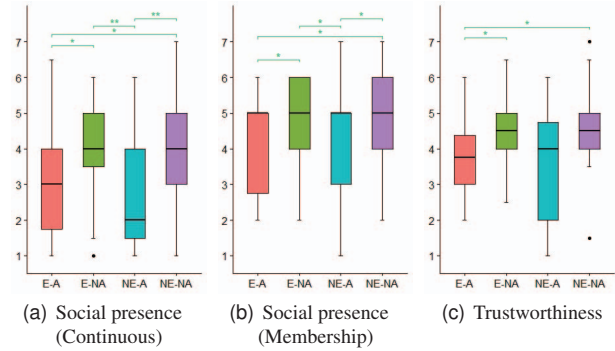


Figure 2: Self-reported results of social presence and trustworthiness (* $p < .05$; ** $p < .01$; and *** $p < .001$).

that the engagement posture can enhance the user's perception of the social presence of the DA. However, in a situation involving group decision-making, affective posture can significantly impair the user's experience. We have no doubt that the presence of DAs within an organization will become commonplace in the near future. We will further explore changes in user experience, especially user behavior, depending on diverse factors of DAs within the organization.

Table 1: Questionnaire was answered with a 7-point Likert scale.

Social Presence (Continuous)

C1 I wanted to continue the conversation with DA.

C2 I wanted to form a relationship with DA.

Social Presence (Membership)

M1 During the discussion, I was looking at and speaking to the DA.

M2 During the discussion, I thought DA as one member of a three-members group

Trustworthiness

T1 I doubt the intentions, thoughts, and conversations of DA.

T2 Whether or not DA's opinion is right, I think the way she talks and acts has made her more sure of what she says.

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